www.tntextbooks.in

$$2C_6H_{12}O_6 + 3O_2 \longrightarrow 3C_4H_6O_5 + 3H_2O + \text{Energy}$$
Glucose — Malic acid

$$\text{RQ of glucose in succulents} = \frac{\text{zero molecule of } CO_2}{\text{3 molecules of } O_2}$$

$$= 0 \text{ (zero)}$$

4. When respiratory substrate is protein or fat, then RQ will be less than unity.

$$2(C_{51}H_{98}O_6) + 145O_2 \longrightarrow 102CO_2\uparrow + 98H_2O + \text{Energy}$$
Tripalmitin(Fat)

$$\text{RQ of Tripalmitin} = \frac{\text{102 molecules of } CO_2}{\text{145 molecules of } O_2}$$

$$= 0.7 \text{ (less than unity)}$$

5. When respiratory substrate is an organic acid the value of RQ will be more than unity.

$$C_4H_6O_5 + 3O_2 \longrightarrow 4CO_2\uparrow + 3H_2O + \text{Energy}$$
Malic acid

$$\text{RQ of malic acid} = \frac{\text{4 molecules of } CO_2}{\text{3 molecules of } O_2}$$

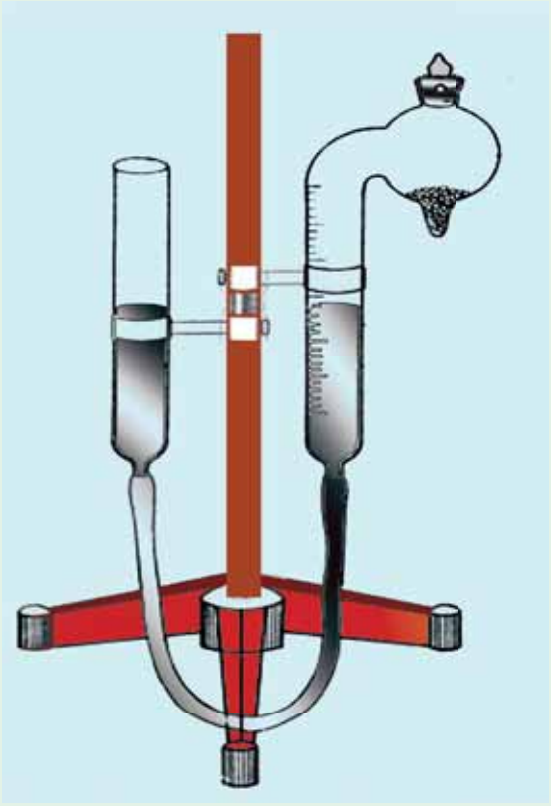
$$= 1.33 \text{ (more than unity)}$$

Significance of RQ

1. RQ value indicates which type of respiration occurs in living cells, either aerobic or anaerobic.
2. It also helps to know which type of respiratory substrate is involved.



Red colour in various parts of plants is due to the presence of anthocyanin, synthesis of which require more O_2 than CO_2 evolved. RQ will be less than one.



The apparatus used for determining respiration and RQ is called Ganong's Respirometer.

Respiratory quotients of some other substances

Proteins	:	0.8–0.9
Oleic acid (Fat)	:	0.71
Palmitic acid (Fat)	:	0.36
Tartaric acid	:	1.6
Oxalic acid	:	4.0

Experiment to demonstrate the production of CO_2 in aerobic respiration

Take small quantity of any seed (groundnut or bean seeds) and allow them to germinate by imbibing them. While they are germinating place them in a conical flask. A small glass tube containing 4 ml of freshly prepared